Lixun (Alex) Zhang

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EDUCATION

Duke University Durham, NC
M.S. in Computer Science Expected: May 2027
GPA: 3.93/4.0

Wake Forest University Winston-Salem, NC
B.S. in Computer Science with Honor and Statistical Science Aug. 2021 – May 2025
GPA: 3.87/4.0 | *cum laude*

RESEARCH EXPERIENCE

Duke University Durham, NC
Advisors: Prof. Yiran Chen, Prof. Hai "Helen" Li Oct. 2025–Present

- Proposed a low-rank KV-cache compression method for long-context LLM inference, using shared Value latents to reduce tensor-parallel communication and guide sparse Key routing.
- Built a standardized evaluation platform for SVD-based LLM compression and investigated cross-task shared subspaces in KV-cached decode-time representations.

Wake Forest University Winston-Salem, NC
Advisor: Prof. Grey Ballard May 2023–Mar. 2025

- Developed an implicit pairwise QR-based alternating least squares method for tensor CP rounding, reducing time complexity from exponential to linear in tensor order while preserving numerical stability.

PUBLICATIONS

[*]Equal contribution

- Z. Shao[*], **L. Zhang**[*], K. Cui[*], W. Wu[*], J. Kim, Y. Wang, T. Jiang, H. Ye, Q. Wang, F. Yang, D. Zhuo, Y. Chen, H. Li, "*LowRankArena: A Standardized Evaluation Platform for SVD-Based LLM Compression*," Conference on Neural Information Processing Systems (NeurIPS), Evaluations and Datasets Track, 2026.
- Z. Shao, **L. Zhang**, K. Cui, Y. Wang, T. Jiang, H. Ye, Q. Wang, Z. Du, Y. Fu, F. Yang, D. Zhuo, Y. Chen, H. H. Li, "*DecodeShare: Tracing the Shared Subspace of LLM Decode-Time Decisions*," International Conference on Machine Learning (ICML), 2026 (**Spotlight, top 2.2%**).
- **A. Zhang**, B. D. Verma, J. Van Lent, G. Ballard, "*Efficient CP Rounding Using Alternating Least Squares with QR Decomposition*," SIAM Journal on Matrix Analysis and Applications (SIMAX), 2026.

PREPRINTS

- **L. Zhang**, Z. Shao, K. Cui, Y. Liang, F. Yang, Y. Chen, "*BasisKV: Value Latents for Communication and Key Routing*," arXiv preprint arXiv:2611.02583, 2026.
- Z. Shao, L. Tian, G. Zemlevskiy, K. Cui, **L. Zhang**, Y. Wang, T. Jiang, J. Kim, Y. Chen, R.-F. Wang, F. Yang, H. Li, Y. Chen, "*The Functional Structure of Post-Compression Recovery in Low-Rank LLMs*," arXiv preprint arXiv:2610.05504, 2026.

SELECTED PROJECTS

GPU-Accelerated Polynomial Arithmetic for Zero-Knowledge Proofs Fall 2026

- Implemented GPU-accelerated number-theoretic transforms using radix-16 decomposition and Monarch-style block factorizations.
- Developed an RNS-based GPU pipeline for low-degree extension, verifying exact results against a CPU reference on over 35 million field elements.

Sparsity-Structure Preserving Tucker Decomposition Apr. 2024–Mar. 2025

- Designed a Kronecker-sketching algorithm using sparse tensor–matrix products to construct sparsity-preserving Tucker cores.
- Implemented a chunked HOSVD baseline to evaluate approximation error against dense-core decomposition methods.

ACADEMIC SERVICE

Reviewer: ICLR 2027

Teaching Assistant Duke University

ECE 661: Computer Engineering Machine Learning and Deep Neural Nets, Fall 2026
COMPSCI 330: Design and Analysis of Algorithms, Fall 2025 & Spring 2026

PRESENTATION

Wake Forest Undergraduate Honor Thesis Presentation | Winston-Salem, NC Apr. 2025

- Presented Efficient CP Rounding Using Alternating Least Squares with QR Decomposition as a partial fulfillment for graduating with honor in Computer Science

Wake Forest Undergraduate Research Day | Winston-Salem, NC Sep. 2024

- Presented a poster on Sparsity-Structure Preserving Tucker Decomposition based on the work done during summer 2024

HONORS & AWARDS

Dean's List, Wake Forest University All Semesters
Upsilon Pi Epsilon, Computer Science Honor Society 2025
Wake Forest Research Fellowship 2024

SKILLS

Programming Languages: *Python, C++, CUDA, MATLAB, R, Java*
Frameworks & Tools: *PyTorch, vLLM, NCCL, Git, Linux*